Dr. Z. Radovilsky
BAN 673 Time Series Analytics:
Summary of Process Steps

Step	*Name*	*Main Decisions*	*Comments*
1	Define Goal	• **Descriptive and/or predictive goal** • **Forecasting horizon** (long-term vs. short-term). • **Forecasting usage** (stakeholders and periodic adjustments). • **Forecast expertise** (in-house or outsourcing; how many and how often to forecast; software for time series forecasting).	• Forecasting is a combination of science and art. • Popular software in business time series forecasting: **Excel** (for simple time series forecasting), **R** (for simple and advanced forecasting methods), **Python** (for simple and advanced forecasting methods), **end-user general statistical software** (e.g., JMP, Stata, SPSS, Minitab), **professional forecasting software** including enterprise resource planning (ERP) systems (e.g., Oracle, SAP, Microsoft, etc.)
2	Get Data	• **Data quality** • **Temporal frequency** • **Series granularity** • **Domain expertise**	• In business, typical frequencies of historical data are monthly, quarterly, daily, weekly, and yearly. • **Minimum data requirements in time series analytics**: at least 36 monthly periods (3 years) for monthly data; at least 16 quarterly periods (4 years) for quarterly data; at least 104 weekly periods (2 years) of weekly data. • Sometimes, it is very useful to aggregate the data and then forecast it to achieve better accuracy results.
3	Explore & Visualize Series	• **Summary statistics of historical time series** (mean, median, standard deviation, etc.). • **Time series patterns (components):** Level (Stationary), Trend, Seasonality, and Noise (Randomness). • **Additive vs. multiplicative components** (specifically, for seasonality). • **Autocorrelation** in various lags. • **Visualize series** with a data plot to check for data series patterns (components).	• Not all time series have trend and/or seasonality, but **all contain level and noise (randomness).** • **Significant autocorrelation coefficients**: lag 1 --> trend; lag 4 -> seasonality in quarterly data; lag 12 -> seasonality in monthly data; autocorrelation coefficients decline rapidly to being insignificant (0) after 2^{nd} or 3^{rd} lags -> level in time series data. • To remove trend and/or seasonality from time series apply differencing method. • For **data predictability** evaluation: **Approach 1** -> hypothesis testing of the autoregression coefficient in *AR(1)* model, *Ho*: β = 1. If *Ho* is rejected (p-value is <= 0.05), this time series is predictable, otherwise – it is random walk; **Approach 2** -> consider autocorrelation coefficients for first differenced data. If the balk

		• **Evaluate data predictability** (time series is 'random walk' -> forecasting methods other than naïve forecast may not be useful; time series is not 'random walk' -> various forecasting methods may be used)	portion of them is random, specifically in lags 1-4 and seasonality lags, this is random walk; otherwise – time series is predictable.
4	Pre-Process Data	• **Detect and clean potential issues in time series data**: outliers, missing values, unequal spacing, irrelevant periods, etc.	• Apply domain expertise to remove or keep outliers in time series historical data. • Utilize a standard imputation approach to replace missing values (similar to Data Mining/Machine Learning imputation). • If unequal spacing is expected in a data set (e.g., in daily stock market data), consider using the data frequency of 1. • Remove records with irrelevant periods.
5	Partition Time Series	• **Training partition (period)** is used for model development. • **Validation (test) partition (period)** is used for model testing.	• **Nonrandom selection** of partition period. • **Early period** in historical data -> **training partition** with 70-85% of historical data. • **Later (more recent) period** in historical data -> **validation partition** with 15-30% of historical data. • The smaller the time series, the less data will go into validation period. • **Recombine partitions to the entire data set for the final forecasting model development** and forecasting into the future. • If data patterns in the validation period are significantly different from those in the training period, the data partitioning may not be useful --> apply the entire data set for model development and forecasting.
6	Apply Forecasting Method(s)	• **Model-driven methods** (e.g., linear/non-linear regression, autoregressive models, ARIMA) • **Data-driven methods** (e.g., naïve and seasonal naïve trailing and weighted moving average, simple and advanced exponential smoothing). • **Two-level forecasting** (e.g., Level 1 -> regression model with trend and seasonality	• Always **select several alternative models** (at least 2-3) for development in training period and performance comparison in the validation period. • Examples of alternative models for time series with **level component only**: naïve, trailing MA, simple exponential smoothing (SES), ARIMA, and auto ARIMA. • Examples of alternative models for times series with **level and trend components only**: naïve, regression with linear/nonlinear trend, Holt's model, AR, non-seasonal ARIMA, auto ARIMA, two-level, and ensembles.

		or Holt-Winter's model; Level 2 -> trailing MA or simple exponential smoothing for residuals of Level 1 forecast; Two-level forecast -> combination of Level 1 and 2 forecasts). • **Ensemble methods** (resulting forecast is the average of several forecasting methods in respective periods; can be weighted average with specific forecast's weight proportional method performance). • **Additional methods**	• Examples of alternative models for times series with **level and seasonality components only**: seasonal naïve, regression with seasonality, Holt-Winter's, AR, seasonal ARIMA, auto ARIMA, two-level, and ensembles. • Examples of alternative methods for times series with **level, trend, and seasonality components**: naïve, seasonal naïve, regression with linear/nonlinear trend and seasonality, Holt-Winter's, AR, seasonal ARIMA, auto ARIMA, two-level, and ensembles. • Develop each model using training partition (period) and test it in validation partition (period). • For the final forecast into the future, **use the entire data** set to develop the model or several alternative models. • **For additional time series methods** consider Hyndman, R.J. & Athanasopoulos, G. Forecasting: Principles and Practice, 2nd Ed., https://otexts.com/fpp2/.
7	Evaluate & Compare Performance	• **Common accuracy performance measures** (ME, RMSE, MAE, MPE, MAPE, MASE, ACF1, Theil's U). • Used to **select the best model for forecasting**. • **Prediction (confidence) intervals** to identify forecast range in the forecasting period.	• The lower the value of each accuracy measure, the more accurate the associated model. • Usually, compare performance measures in validation period and/or entire data set. • Utilize naïve and/or seasonal naïve forecast as a **baseline for comparison**. The identified best model(s) should have more accurate performance measures than the baseline forecast(s). • **MAPE and RMSE should be considered as the preferred performance measures** for model selection, with MAPE being of higher importance than RMSE in business time series forecasting. • Typical prediction intervals are based on 95% confidence level probability (in R, default to 80% and 95% confidence level).
8	Implement Forecast/ System	• Incorporate forecast into the existing enterprise IT system. • **Create and maintain IT support system**. • **Periodic review and reevaluation of forecast**.	• As new historical data becomes available, consider reevaluating the forecasting model. • The **reevaluation interval should be approximately every 3-6 months** (from one quarter to semi-annual) and should include steps 2-8 of the time series analytics process.